

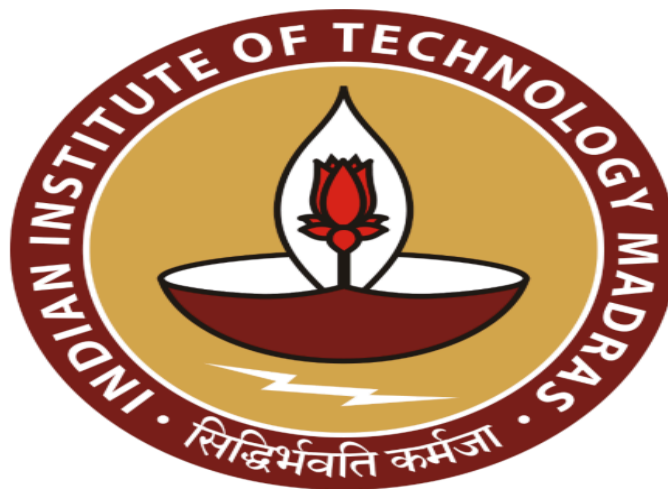
Enhancing Profitability and Cash Flow at a Grocery Store: A Data-Driven Approach to Credit, Financial, and Margin Management

A Mid-term report for the BDM capstone Project

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1 Executive Summary

Faiz General Store, established in 1995 and formerly known as Aslam General Store, is a family-owned retail business generating approximately ₹30 lakhs in annual revenue, primarily managed by two family members with occasional support from extended family. The business faces three critical challenges: unmanaged credit sales constituting 35% of revenue, suboptimal profit margins averaging 12.5%, and limited analytical capabilities for strategic decision-making.

The analysis leverages primary data collected across three domains. Credit management variables (date, credit amount, payment dates, outstanding balances) enable customer risk assessment. Category performance metrics (date, category, revenue, cost) facilitate product portfolio optimization. Financial health indicators (daily revenue, cost, cash/credit sales) provide business stability insights.

Preliminary analysis reveals significant opportunities for improvement. A rule-based credit rating system effectively segments customers into loyalty categories, addressing the 9% default rate by identifying high-risk patrons. Category-wise analysis uncovers misaligned focus areas, notably revealing that high-margin categories like Personal Care receive less attention than low-margin Staples. Financial metrics demonstrate liquidity patterns and profitability trends essential for cash flow management.

The implemented methodologies—prioritizing simplicity and business applicability—deliver actionable insights for credit management, category optimization, and financial monitoring. These data-driven approaches empower the business owner with strategic decision-making capabilities previously unavailable, directly addressing the core challenges of profitability, risk management, and strategic focus.

2 Proof of originality

This master folder contains all the things -  BDM_Project_Proof_of_Originality_23F2003821

3 Metadata and Descriptive Statistics

All the sheets (Raw and Analysis with charts with proper name mention) are in this complete file -  Faiz General Store -Data

The metadata and descriptive statistics will be described by problem statement-wise for better understanding, interpretability and readability.

3.1 Customer credit rating

For this the data was collected from shop owners credit register from 1 march 2025- 31 august 2025 and manually entered in google sheet. The sheet named “Raw- credit” has 2625 rows and 9 columns.

Metadata:

Table 1: This table provides an overview of the 9 columns in the credit dataset, including the column name, data type, and a brief description of each variable and data sources.

Column Name	Data Type	Description & Relevance
Date	Date	Date when credit was given — used to calculate repayment delay.
Customer_name	String	Identifies customers for credit behavior classification.
Credit_amt	Integer	Total credit amount (₹) — indicates financial exposure.
Payment_date	Date	Used to calculate delay in repayment.
Amount_repaid	Integer	Amount paid (₹) — helps identify full or partial payments.
Outstanding	Integer	Remaining dues (₹) — key for credit risk.
Payment_status	String	Repayment status (full, partial, pending).
Outstanding_status	String	Whether dues are finally cleared.

Outstanding_clearing _delay	Integer	Days taken to clear dues — key delay measure.
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Descriptive Statistics:

Table 2: This table summarizes the descriptive statistics of selected variables from the Raw-credit dataset.

Statistic	Credit_amt	Amount_repaid	Outstanding	Outstanding_clearing_delay
Count	2621	2298	2265	1852
Mean	206.51	119.28	117.95	27.70
Std Dev	97.87	86.35	83.47	13.92
Min	38	10	10	1
25%	122	49	48	16
50%(Median)	208	98	101	26
75%	288	169	170	39
Max	378	376	377	58

Justification - The descriptive statistics reveal critical credit risk insights: average delay, credit amount range, and variability metrics directly address unmanaged credit sales by enabling data-driven credit limits and customer risk segmentation.

3.2 Category-wise Analysis

For this problem statement, a printed and spiral binded copy was given to the shop owner, and he entered daily category-wise revenue and cost in it. The data manually entered in google sheet. The data is collected from 1 March 2025 - 31 August 2025. The sheet named Raw-Category-wise has 1105 rows and 4 columns.

Metadata:

Table 3: This table provides an overview of the 4 columns in the Raw-category-wise dataset, including the column name, data type, and a brief description of each variable and data sources.

Column Name	Data Type	Description & Relevance
Date	Date	Date of sales — used to analyze category performance.
Category	String	Product category name — basis for comparing category-wise performance.
Revenue	Integer	Total daily revenue (₹) — key metric for assessing category contribution.
Cost	Integer	Total daily cost (₹) - used to calculate profit and profit margin for categories.

Descriptive Statistics:

Table 4: This table summarizes the descriptive statistics of selected variables from the Raw-category-wise dataset.

Category	Revenue Mean	Revenue Std	Revenue median	Cost Mean	Cost Std	Cost median
Staples	5005.18	1800.89	970.5	4532.52	1647.33	829.5
Beverages	1050.32	510.36	973	893.61	431.78	829
Personal Care	323.71	440.15	854	235.88	386.19	731
Dairy and Bakery	981.22	380.91	970	849.53	324.15	829
Snacks	441.33	1708.79	879	346	1562.47	740
Cleaning & Household	931.55	408.36	959	783.24	345.57	812

Justification: Category-wise descriptive statistics (average revenue, cost, standard deviation, median) directly solve the limited visibility problem by revealing performance patterns and stability across segments.

- Business Impact: Identifies focus areas for management and promotional strategy.

3.3 Financial Health Analysis

For this problem, the data was collected from the shop owner’s daily summary register from Sep 2024 - Aug 2025, and manually entered in google sheet.The sheet named Raw-Day-wise has 366 Rows and 5 columns.

Metadata:

Table 5: This table provides an overview of the 5 columns in the Raw-Day-wise dataset, including the column name, data type, and a brief description of each variable and data sources.

Column Name	Data Type	Description & Relevance
Date	Date	Date of the transaction. Essential for analyzing revenue and profit trends over time.
Revenue	Integer	Total daily sales revenue (INR). Primary measure of store performance and financial health.
Cash_Sale	Integer	Portion of revenue from cash sales (INR). Crucial for calculating cash-to-credit ratio and liquidity risk.
Credit_Sale	Integer	Portion of revenue from credit sales (INR). Key to understanding reliance on credit affecting cash flow.
Cost	Integer	Total daily cost of goods sold (INR). Basis for profitability calculations and gross profit analysis.

Descriptive Statistics:

Table 6: This table summarizes the descriptive statistics of selected variables from the Raw-Day-wise dataset.

Statistic	Revenue	Cash_Sale	Credit_Sale	Cost
Count	365	365	365	365
Mean	9000	6181.72	2818.25	7856.91
Std Dev	2192.20	1950.21	1031.69	1907.31

Min	0	0	0	0
25%	7706	4942	2118	6698
50%(Median)	8756	5916	2786	7679
75%	10000	7161	3532	8677
Max	18646	16528	5557	16390

Justification: Financial metrics (such as revenue consistency with Std: 2,192.20 daily average of ₹9,000) are directly related to business stability, liquidity assessment, and cash flow management.

4 Detailed Explanation of Analysis Process/Method

4.1 Data Cleaning and Preprocessing:

The Raw-Day-wise and Raw-Category-wise data found no anomalies and its quantitative so no special cleaning needed, for total 5 closing days because the shop was closed for famous festivals so people know it before many days so they bought all items before the closing day so for perfect normalization 0 is filled in the blank space. The Raw-Credit dataset also found no major anomaly but 370 rows have missing values in Payment date, and shop owner unable to provide that dates, but out of that 322 rows has outstanding so for delay days their Outstanding_clearing_delay used but for remaining 48 rows, detailed dataset analyzed and its concluded to fill that values with dates so their delay days should be average of delay days which is 15 days. In Payment_status for partial payment two values filled with two different status - part, partial, so all the partial payment status updated with partial status. All the values are cross-verified for perfect analysis.

4.2 Analysis Process/Method

Google Colab Notebook - [BDM_Project_Credit_Rating_Analysis.ipynb](#)

This method is also described problem statement-wise for perfect readability and interpretability.

4.2.1 Customer Credit Rating: For this solution firstly the Delay_days column was derived using the rule if Amount_repaid > Outstanding then Delay_days = Payment_date - Date, if Amount_repaid < Outstanding then Delay_days = Outstanding_clearing_delay. After that a

summary sheet is created with columns - Customer_name, Total_credit, Total_repaid, Total_defaulted, Average_delay, default_percentage(derived using Total_defaulted/Total_credit) group by Customer_name. Then rule based credit rating method performed and credit rating from S, A, B, C, D, E, F is assigned to customers:

S: If default percentage 0, and average_delay $\leq$ 20 days.

A: If default percentage 0, and average_delay $>$ 20 days.

B: If $0 <$ default percentage $\leq$ 10 and average_delay $\leq$ 20 days.

C: If $0 <$ default percentage $\leq$ 10 and average_delay $>$ 20 days.

D: If $10 <$ default percentage $\leq$ 20 and average_delay $\leq$ 30 days.

E: If $10 <$ default percentage $\leq$ 20 and average_delay $>$ 30 days.

F: If default percentage $>$ 20.

All this rule applied on Google Colab using python and credit rating assigned to customers.

Justification: Since our focus is on business value rather than complex technicality. The dataset is very limited and has only 105 customers, so machine learning is not giving good results more than rule based. If its supermarket data where 500 plus customers then there ML is best but here with almost fixed customers and not very new customers adding because its a local shop this rule based is perfect for both accuracy and shop owner understanding and use cases. So this is the best method for the shop. But before final many machine learning algorithms will be checked and if any of them outperform this, then will replace, but 90% chances that this rule based rating outperform machine learning algorithms on this limited dataset.

4.2.2 Category-wise Analysis: For this two new columns derived- Profit using formula Revenue- Cost and Profit Margin using formula $\text{=ROUND}((\text{Profit}/\text{Revenue}) * 100, 2)$. Then categorywise revenue and margin analysed in detail using descriptive statistics and visualization.

Justification: This method is best in both recommendation and feasibility. It tells us everything in detail like category-wise revenue distribution, margin distribution. It gives us complete information like which category is high revenue low margin, which is low revenue high margin etc, so it gives complete information to focus on which category, which category to promote.

4.2.3 Financial Health Analysis: For this two new columns derived - Profit margin using

formula $\text{=ROUND}((\text{Profit}/\text{Revenue}) * 100, 2)$ and Cash_to_credit_ratio using formula $\text{=ROUND}((\text{Cash_sale}/\text{Credit_sale}) * 100, 2)$. After that using line area chart cash to credit ratio visualize and trendline tells the business liquidity. Using bar chart revenue is visualized to assess the business stability.

Justification: This approach provides a complete snapshot of business liquidity, stability and cash flow. It is easy for the shop owner to understand and not overly complex, so it's the best approach here in this context.

Strategic Note: All analysis methods were developed after a detailed understanding of the business domain and needs. Each method prioritizes simplicity, business value, and real-world usability over technical complexity. The focus remains on providing insights that the business can truly use and appreciate.

5 Results and Findings

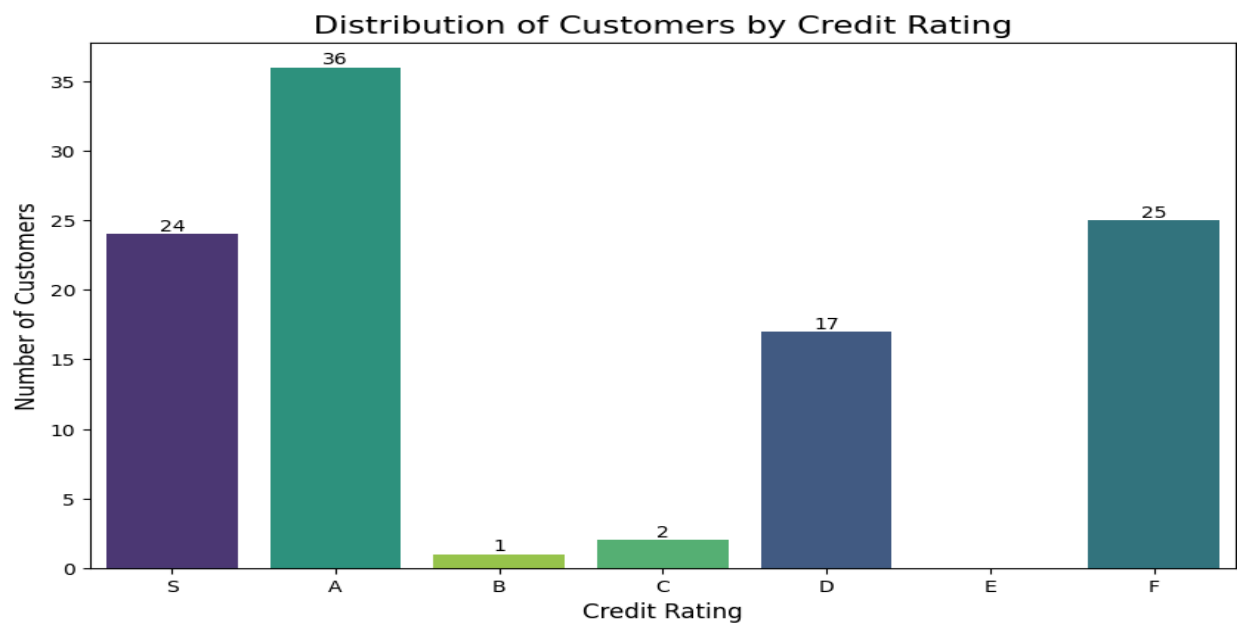


Figure 1: Distribution of Customers by Credit Rating

The analysis of customer credit ratings, shown in Figure 1, reveals a highly polarized customer base. A significant majority of customers are clustered at the extremes, with 60 customers in the low-risk 'S' and 'A' categories and 42 customers in the high-risk 'D' and 'F' categories. The low

count of customers in the intermediate 'B' and 'C' ratings is a crucial finding. This suggests that for this business, customers do not typically remain in a moderately risky state; their payment behavior tends to be either reliable or it deteriorates significantly. For the shop owner, this underlines the dual importance of nurturing their large base of good customers while closely managing the substantial financial risk posed by the large high-risk segment.

Category-wise Revenue Contribution vs Average Profit Margin

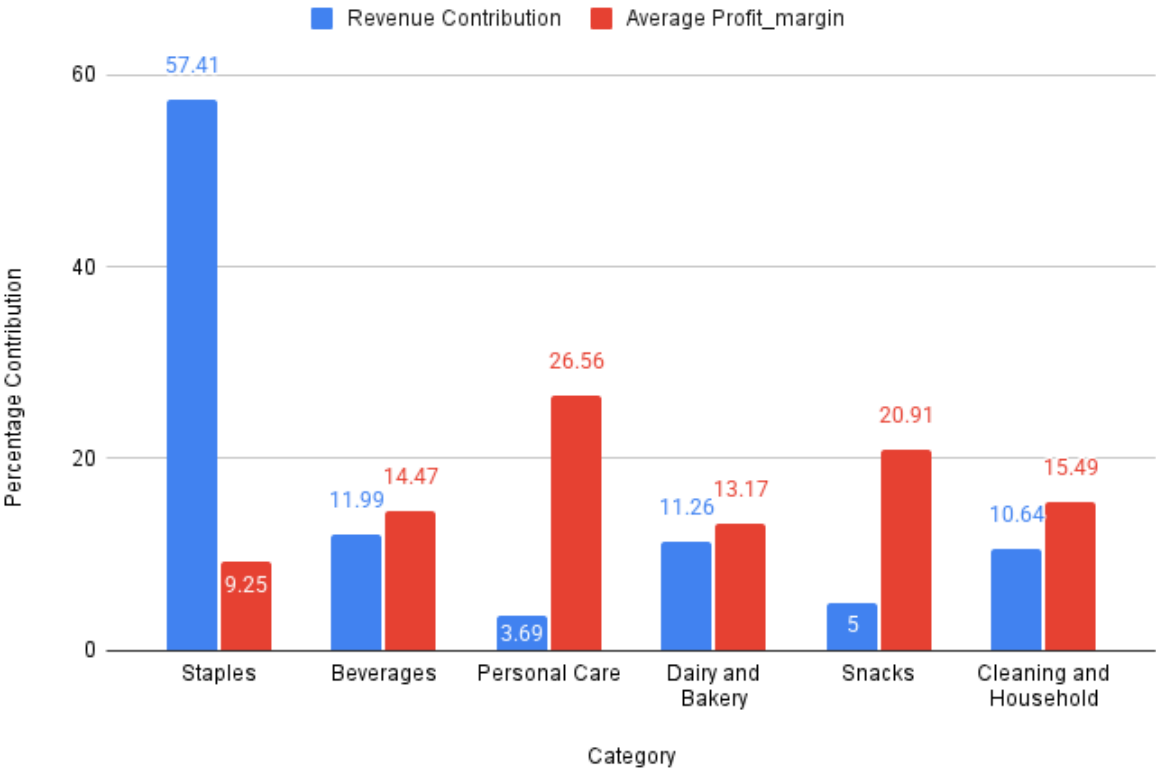


Figure 2: Category-wise Revenue Contribution vs. Average Profit Margin

This analysis reveals a clear trade-off between sales volume and profitability. The **Staples** category drives the majority of revenue (57.41%) but operates on the lowest average profit margin (9.25%). Conversely, the most profitable categories, like **Personal Care** (26.56% margin) and **Snacks** (20.91% margin), contribute the least to total sales.

This finding is critical: while staples are essential for attracting customers, the shop's overall profitability is highly dependent on increasing the sales of these high-margin goods through strategic promotions or better placement.

Cash_to_Credit_Ratio Trend

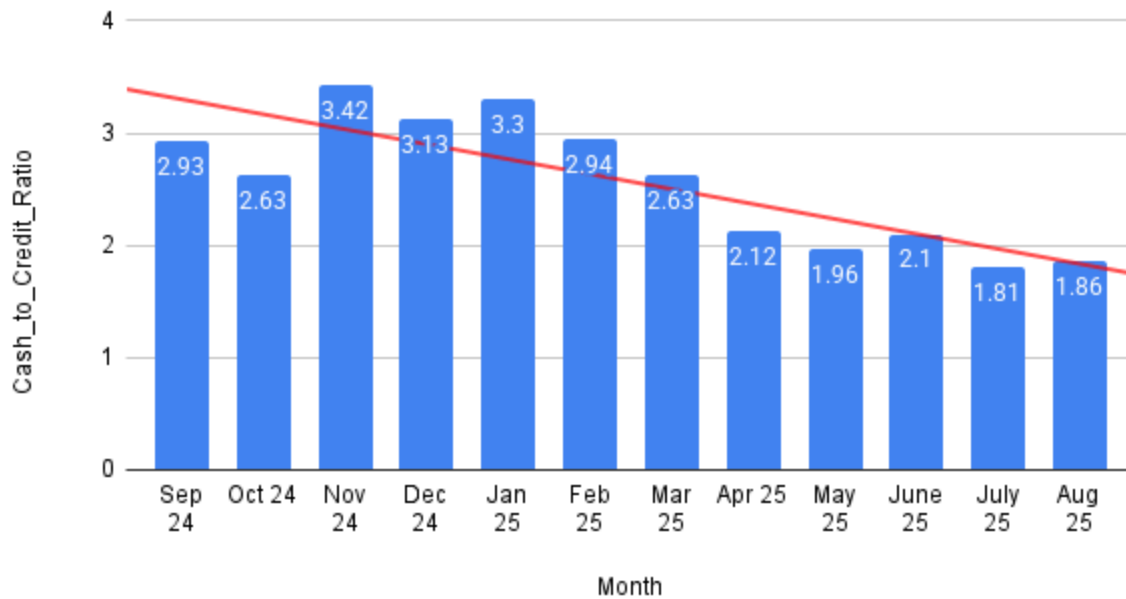


Figure 3: Cash-to-Credit Ratio Trend

This chart tracks the ratio of cash-to-credit sales over twelve months, revealing a significant downward trend. The ratio has fallen from a peak of 3.42 to consistently below 2.0, indicating the business is becoming increasingly reliant on credit sales. This shift represents a critical financial risk, as it negatively impacts the shop's immediate **cash flow** and increases its exposure to bad debt from non-paying customers.

Note:

Figure 1 chart is on google colab and link provided above.

Figure 2 chart is on google sheet and tab name - Analysis-Categorywise

Figure 3 chart is on google sheet and tab name - Analysis-Daywise

Complete file once more for reference - [📁 Faiz General Store -Data](#)